

Linear System Theory

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Transfer Function

- For linear time-invariant systems, transfer functions provide another way of describing input-output systems
 - State space description: Time-domain
 - Transfer function: Frequency-domain
- To discuss transfer functions of continuous-time systems, we need the *Laplace transform*

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1. 1	$\frac{1}{s}$	2. e^{at}	$\frac{1}{s-a}$
3. $t^n, n=1,2,3,\dots$	$\frac{n!}{s^{n+1}}$	4. $t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5. $\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{3/2}}$	6. $t^{n-1/2}, n=1,2,3,\dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+1/2}}$
7. $\sin(at)$	$\frac{a}{s^2 + a^2}$	8. $\cos(at)$	$\frac{s}{s^2 + a^2}$
9. $t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$	10. $t \cos(at)$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
11. $\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$	12. $\sin(at) + at \cos(at)$	$\frac{2as^2}{(s^2 + a^2)^2}$
13. $\cos(at) - at \sin(at)$	$\frac{s(s^2 - a^2)}{(s^2 + a^2)^2}$	14. $\cos(at) + at \sin(at)$	$\frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$
15. $\sin(at + b)$	$\frac{s \sin(b) + a \cos(b)}{s^2 + a^2}$	16. $\cos(at + b)$	$\frac{s \cos(b) - a \sin(b)}{s^2 + a^2}$
17. $\sinh(at)$	$\frac{a}{s^2 - a^2}$	18. $\cosh(at)$	$\frac{s}{s^2 - a^2}$
19. $e^{at} \sin(bt)$	$\frac{b}{(s-a)^2 + b^2}$	20. $e^{at} \cos(bt)$	$\frac{s-a}{(s-a)^2 + b^2}$
21. $e^{at} \sinh(bt)$	$\frac{b}{(s-a)^2 - b^2}$	22. $e^{at} \cosh(bt)$	$\frac{s-a}{(s-a)^2 - b^2}$
23. $t^n e^{at}, n=1,2,3,\dots$	$\frac{n!}{(s-a)^{n+1}}$	24. $f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
25. $u_c(t) = u(t-c)$ Heaviside Function	$\frac{e^{-cs}}{s}$	26. $\delta(t-c)$ Dirac Delta Function	e^{-cs}
27. $u_c(t) f(t-c)$	$e^{-cs} F(s)$	28. $u_c(t) g(t)$	$e^{-cs} \mathcal{L}\{g(t+c)\}$
29. $e^{at} f(t)$	$F(s-c)$	30. $t^n f(t), n=1,2,3,\dots$	$(-1)^n F^{(n)}(s)$
31. $\frac{1}{t} f(t)$	$\int_s^\infty F(u) du$	32. $\int_0^t f(v) dv$	$\frac{F(s)}{s}$
33. $\int_0^t f(t-\tau) g(\tau) d\tau$	$F(s)G(s)$	34. $f(t+T) = f(t)$	$\frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$
35. $f'(t)$	$sF(s) - f(0)$	36. $f''(t)$	$s^2 F(s) - sf'(0) - f''(0)$
37. $f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$		

Transfer Function

Consider now the LTI system

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

along with the initial condition

$$x(0) = x_0 .$$

Taking the Laplace transform

$$s\hat{x}(s) - x_0 = A\hat{x}(s) + B\hat{u}(s) \quad \rightarrow \quad (sI - A)\hat{x}(s) = x_0 + B\hat{u}(s)$$

$$\hat{y}(s) = C\hat{x}(s) + D\hat{u}(s) \quad \rightarrow \quad \hat{y}(s) = C(sI - A)^{-1}x_0 + \hat{G}(s)\hat{u}(s)$$

The **Transfer function (matrix)** is defined by

$$\boxed{\hat{G}(s) = C(sI - A)^{-1}B + D}$$

Zero initial conditions response

- Suppose that $x(0) = 0$. Then the relationship between the Laplace transform of the input and the output is given by a simple relationship

$$\hat{y} = \hat{G}\hat{u}$$

- For a SISO system, for $u \neq 0$,

$$\hat{G} = \frac{\hat{y}}{\hat{u}}.$$

Rational transfer functions

- A scalar valued function of a complex variable is rational if it can be expressed as the ratio of two polynomials
- **Proper**: the order of the numerator polynomial $\leq$ the order of the denominator polynomial
- **Strictly Proper**: the order of the numerator polynomial $<$ that of the denominator polynomial

General LTI input output system

- $\hat{G}(s) = C(sI - A)^{-1}B + D = \frac{1}{\Delta(s)} N(s)$
 - $\Delta(s) = \det(sI - A)$: characteristic polynomial of A
 - $\deg N(s) \leq \deg \Delta(s)$
- Note that: state space to TF is unique, but TF to state space might not be unique

Discrete time transfer function

Consider a discrete time LTI system

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned}$$

- The transfer function matrix is defined by

$$\hat{G}(z) = C(zI - A)^{-1}B + D$$

where $z \in \mathbb{C}$. This is the z -transform of the impulse response matrix G .

State space realization of transfer functions

- Given a matrix valued function $\hat{G}$ of a complex variable s , find constant matrices A, B, C, D (or determine that none exist) so that

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

* $\hat{G}$ has a finite-dimensional
state space realization
if and only if
 $\hat{G}$ is rational &
proper

Minimal realization

- A given transfer function has an infinite number of state space realizations. One might expect that the dimension of every state space realization is the same. This is not even true as the next example illustrates. We say that a realization is a minimal realization if its dimension is less than or equal to that of any other realization. Clearly, all minimal realizations have the same dimension.

$$A = \begin{pmatrix} 1 & -3 \\ -3 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$C = (1 \ 0) \quad D = 0$$

$$\begin{aligned} \hat{G}(s) &= C(sI - A)^{-1}B + D \\ &= \frac{s-4}{s^2-2s-8} \\ &= \frac{1}{s+2} \end{aligned}$$

$$A = -2 \quad B = 1 \quad C = 1 \quad D = 0$$

Realization of SISO transfer functions

$$\hat{G}(s) = \frac{\beta_{n-1}s^{n-1} + \dots + \beta_1 s + \beta_0}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1 s + \alpha_0} + d$$

- Controllable canonical form realization

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \dots & -\alpha_{n-2} & -\alpha_{n-1} \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

- Observable canonical form realization

$$A = \begin{pmatrix} -\alpha_{n-1} & 1 & 0 & \dots & 0 & 0 \\ -\alpha_{n-2} & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -\alpha_1 & 0 & 0 & \dots & 0 & 1 \\ -\alpha_0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix} \quad C = [\beta_0 \quad \beta_1 \quad \dots \quad \beta_{n-2} \quad \beta_{n-1}]$$

$$B = \begin{pmatrix} \beta_{n-1} \\ \beta_{n-2} \\ \vdots \\ \beta_1 \\ \beta_0 \end{pmatrix} \quad D = d$$

$$\hat{G}(s) = \frac{s^2}{s^2 + 1} = \frac{s^2 + 1 - 1}{s^2 + 1} = \frac{-1}{s^2 + 1} + 1$$

$\neq \alpha_0$

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \\ -\alpha_0 & -\alpha_1 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = (-1 \quad 0)$$

$$D = 1$$

$$A = \begin{pmatrix} 0 & 1 \\ -\alpha_1 & -1 \\ -\alpha_0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} \beta_1 \\ \beta_0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$C = (1 \quad 0) \quad D = 1$$



Realization of MIMO transfer function

$$\hat{G}(s) = \begin{bmatrix} \hat{G}_1(s) & \hat{G}_2(s) \\ \hat{G}_3(s) & \hat{G}_4(s) \end{bmatrix}$$

- A proper rational transfer function matrix $\hat{G}(s)$ of size $p \times m$ can be expressed as:

$$\hat{G}(s) = C(sI - A)^{-1}B + D = N(s)\Delta(s)^{-1} + D$$

- Expanding the polynomial matrices $N(s)$ and $\Delta(s)$ in powers of s :

$$N(s) = s^{l-1}N_{l-1} + \dots + sN_1 + N_0$$

$$\Delta(s) = s^l I + s^{l-1}\Delta_{l-1} + \dots + s\Delta_1 + \Delta_0$$

$$N_i = p \times m \text{ matrices}$$

$$\Delta_i = m \times m \text{ matrices}$$

Controllable canonical form realization of MIMO transfer function

- For system dimension $n = m \cdot l$, the controllable realization state-space matrices (A, B, C, D) are defined as:

$$\begin{aligned}
 A &= \begin{pmatrix} 0 & I & 0 & \cdots & 0 & 0 & m \times m \\ 0 & 0 & I & \cdots & 0 & 0 & \\ \vdots & 0 & 0 & \cdots & 0 & I & m \times m \\ 0 & 0 & 0 & \cdots & 0 & I & m \times m \\ -\Delta_0 & -\Delta_1 & -\Delta_2 & \cdots & -\Delta_{q-2} & -\Delta_{q-1} & n \times n \end{pmatrix} \\
 B &= \begin{pmatrix} 0 \\ 0 \\ \vdots \\ I \end{pmatrix}_{n \times m} \\
 C &= (N_0 \ N_1 \ N_2 \ \cdots \ N_{q-2} \ N_{q-1})_{p \times n} \\
 D &= D
 \end{aligned}$$

$\hat{G}(s) = p \times m$ matrices

$$\begin{aligned}
 \frac{N(s)}{\Delta(s)} &= \frac{N_{q-1}s^{q-1} + \cdots + N_0}{s^q + \cdots + 1} \\
 \frac{\Delta(s)}{N(s)} &= \frac{s^q + \cdots + 1}{N_{q-1}s^{q-1} + \cdots + N_0}
 \end{aligned}$$

$$\Delta(s) = s^2 I + \Delta_{p-1} s^{p-1} + \dots + \Delta_1 s + \Delta_0$$

$$sI s^0 : s^2 + \alpha_{p-1} s^{p-1} + \dots + \alpha_1 s + \alpha_0$$

$$\Delta(s) = s^2 I + \alpha_{p-1} I s^{p-1} + \dots + \alpha_1 I s + \alpha_0 I$$

$$A = \begin{pmatrix} 0 & I & 0 & \dots & 0 & 0 \\ 0 & 0 & I & \dots & 0 & 0 \\ \vdots & & & & & \\ -\alpha_1 I & -\alpha_2 I & -\alpha_3 I & \dots & -\alpha_{p-2} I & -\alpha_{p-1} I \end{pmatrix}$$

$m \times m$



Example 36 Consider

$$\hat{G}(s) = \begin{pmatrix} \frac{3}{s-2} \\ \frac{s+4}{s^2-2s+3} \end{pmatrix} \quad \begin{matrix} p \times m \\ 2 \times 1 \end{matrix} \quad \begin{matrix} n = m \times l \\ = 1 \times 3 \end{matrix} \quad A = 3 \times 3$$

$$= \begin{pmatrix} \frac{3(s^2-2s+3)}{(s-2)(s^2-2s+3)} \\ \frac{(s-2)(s+4)}{(s-2)(s^2-2s+3)} \end{pmatrix} = \begin{pmatrix} \frac{3s^2-6s+9}{s^3-s-6} \\ \frac{s^2+2s-8}{s^3-s-6} \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 6 & 1 & 0 & 0 \end{pmatrix}$$

$-\alpha_0 \quad -\alpha_1 \quad -\alpha_2$

$$B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad C = \begin{pmatrix} N_0 & N_1 & N_2 \\ 9 & -6 & 3 \\ -8 & 2 & 1 \end{pmatrix}$$

$$D = 0$$

Example 37 Consider the strictly proper rational function,

$$p = 3 \\ m = 2$$

$$\hat{G}(s) = \begin{pmatrix} \frac{1}{s+1} & \frac{1}{s^2+3s+2} \\ 0 & \frac{1}{s+2} \end{pmatrix} \quad (s+1)(s+2)$$

2×2

$$n = m \times 2$$

$$= 2 \times 2$$

$$A: 4 \times 4$$

$$= \begin{pmatrix} \frac{s+2}{s^2+3s+2} \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{c} \frac{1}{s^2+3s+2} \\ \frac{s+1}{s^2+3s+2} \end{array} \right) \quad \lambda = 2$$

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ -2 & 0 \\ 0 & -2 \end{pmatrix}$$

$-2I_{2 \times 2}$

$$B = \begin{pmatrix} 0 & 1 \\ -3 & 0 \\ 0 & -3 \end{pmatrix}$$

$-2I_{2 \times 2}$

$$C = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$$

$$D = \begin{pmatrix} N_0 & N_1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$D = 0$



Observable canonical form realization of MIMO transfer function

- * Minimal realization is both controllable & observable
- Same system as previous slides shown, the observable canonical form realization can be written as:

$$\hat{G}(s) = N(s)\Delta(s)^{-1} + D$$

$$A = \begin{pmatrix} -\Delta_{n-1} & I & 0 & \dots & 0 & 0 \\ -\Delta_{n-2} & 0 & I & \dots & 0 & 0 \\ \vdots & & & & & \\ -\Delta_1 & 0 & 0 & \dots & 0 & I \\ -\Delta_0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix} = \begin{pmatrix} -\alpha_{n-1}I & I & 0 & \dots & 0 & 0 \\ -\alpha_{n-2}I & 0 & I & \dots & 0 & 0 \\ \vdots & & & & & \\ -\alpha_1I & 0 & 0 & \dots & 0 & I \\ -\alpha_0I & 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} N_{n-1} \\ N_{n-2} \\ \vdots \\ N_1 \\ N_0 \end{pmatrix}$$

$$C = (I \ 0 \ \dots \ 0 \ 0)$$

$$D = D$$

Outline

- Some Linear Algebra



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Linear equations

- Consider the following m linear scalar equations in n scalar unknowns

$$\begin{aligned}x_1, x_2, \dots, x_n: & a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ & a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ & \vdots \\ & a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m\end{aligned}$$

- A basic problem is that of determining whether or not the above system of linear equations has a solution for all x and, if a solution exists, determining all solutions.

Linear equations in matrix/vector form

$$b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\Rightarrow \underline{Ax = b}$$

$$x = n \times 1 \quad b = m \times 1 \quad A = m \times n$$

Applying elementary row operations on the augmented matrix: $(A \quad b)$

- ✓ 1) Interchanging two rows.
- ✓ 2) Multiplication of a row by a nonzero scalar.
- ✓ 3) Addition of a multiple of one row to another different row.

$$\boxed{(A \quad b)}$$



Reduced row echelon

- All zero rows come after the nonzero rows.
- All leading elements are equal to one.
- The leading elements have a staircase pattern.
- If a column contains a leading element of some row, then all other elements of the column are zero.
- Every matrix can be transformed into reduced row echelon form by a sequence of elementary row operations.
- rref in Matlab/python

Reduced row echelon

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{pmatrix}$$

Handwritten annotations: Purple boxes around the leading 1s in each row. Purple arrows point to the first three rows.

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Handwritten annotations: Purple boxes around the leading 1s in rows 1 and 2. Purple arrows point to the first two rows.

$$\begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \end{pmatrix}$$

Handwritten annotations: Red arrow pointing to the zero row. Red curved arrow indicating a row operation.

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Handwritten annotations: Red box around the 2 in row 1, column 2. Red arrow pointing to it. Red text "row 1 - 2 row 2" with an arrow pointing to the operation. Purple boxes around the leading 1s in rows 1 and 2.

- The leading elements of a reduced row echelon matrix are called pivots.

Gauss-Jordan procedure

$$\begin{aligned} y - 2z &= 4 \\ x - y &= 1 \\ 3x - 3y + 4z &= -9 \\ -2y + 10z &= -26 \end{aligned}$$

Here the augmented matrix is given by

$$(A \ b) = \begin{pmatrix} 0 & 1 & -2 & 4 \\ 1 & -1 & 0 & 1 \\ 3 & -3 & 4 & -9 \\ 0 & -2 & 10 & -26 \end{pmatrix}$$

$$x = -1$$

$$y = -2$$

$$z = -3$$

$$\begin{pmatrix} 0 & 1 & -2 & 4 \\ 1 & -1 & 0 & 1 \\ 3 & -3 & 4 & -9 \\ 0 & -2 & 10 & -26 \end{pmatrix} \xrightarrow{r_3 - 3 \times r_1} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & -2 & 4 \\ 3 & -3 & 4 & -9 \\ 0 & -2 & 10 & -26 \end{pmatrix} \xrightarrow{r_3 - 3 \times r_2} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 1 & -12 \\ 0 & -2 & 10 & -26 \end{pmatrix}$$

$$\xrightarrow{r_4 + 2 \times r_2} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 1 & -12 \\ 0 & 0 & 6 & -18 \end{pmatrix} \xrightarrow{\substack{r_2 + 2 \times r_3 \\ r_4 - 6 \times r_3}} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -12 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -12 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$



Linear equations

- A unique solution. (All columns except the last have a leading element.)
- No solution. (The last column has a leading element, that is, there is a row in which all elements are zero except the last element.)
- Infinite number of solutions. (There is a column in $\hat{A}$ which does not have a leading element.)

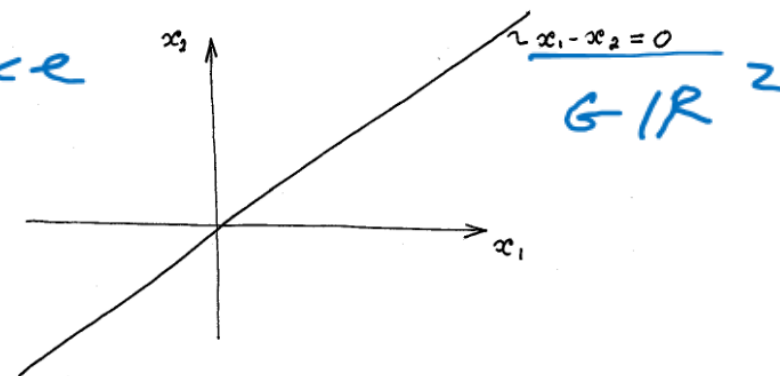
$x_i = \text{anything}$

Subspace

$\mathbb{R}^2 =$ line pass through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
 $\mathbb{R}^3 =$ line / plane pass through $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

- Definition: A non-empty subset S of a vector space is a subspace if for every pair of vectors x, y in S and every pair of scalars α, β , the vector $\alpha x + \beta y$ is contained in S
- A subspace in linear algebra is a subset of a larger vector space that is also a valid vector space under the same operations
 - **Zero vector:** The set must contain the zero vector
 - **Closed under addition:** If you add any two vectors in the set, the result stays in the set.
 - **Closed under scalar multiplication:** If you multiply a vector in the set by any number (scalar), the result stays in the set.

$x_1 - x_2 - 1 = 0$ not a subspace



Span

The span of a bunch X of vectors is the set of all vectors of the form

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m$$

where each x_i is in the bunch & α_i is scalar

$$\mathbb{R}^2 = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

These do not affect span:

- Multiplication of one of the vectors by a nonzero scalar.
- Replacing a vector by the sum of itself and a scalar multiple of another (different) vector in the bunch.
- Removal of a vector which is a linear combination of other vectors in the bunch.
- Inclusion of a vector which is in the span of the original bunch.



Linear independence

A bunch of vectors, $x^1, x^2, \dots, x^m$ is linearly dependent if there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_m$, not all zero, such that

$$\alpha_1 x^1 + \alpha_2 x^2 + \dots + \alpha_m x^m = 0$$

The bunch is linearly independent if it is not linearly dependent, that is, the bunch is linearly independent if and only if (5.4) implies that

$$\alpha_1, \alpha_2, \dots, \alpha_m = 0$$

These do not affect linear independence status:

- Multiplication of one of the vectors by a nonzero scalar.
- Replacing a vector by the sum of itself and a scalar multiple of another (different) vector in the bunch.
- Removal of a vector from the bunch.
- Inclusion of a vector which is not in the span of the original bunch.



$$x_1 \quad x_2$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$2x_1 - x_2 = 0$$

$$\alpha_1, \alpha_2 \neq 0$$

linear dependent

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 7 \\ 6 \\ 5 \end{pmatrix}$$

$$\alpha_1, \alpha_2 = 0$$

linear independent

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 7 \\ 6 \\ 5 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

$$x_1 \quad x_2 \quad x_3$$

$$\alpha_1 = 1$$

$$\alpha_2 = -1$$

$$\alpha_3 = 2$$

$$x_1 - x_2 + 2x_3 = 0$$

linear dependent



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Basis

$$\mathbb{R}^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

- A basis for a vector space is a bunch of **linearly independent** vectors whose **span** is the space.
- A vector space is called *finite dimensional* if it has a *finite basis*; otherwise it is *infinite dimensional*.
- **Dimension:** The number of vectors in any basis of V , denoted $\dim V$.
- An n -dimensional space has n **basis vectors**.
- In an n -dimensional space:
 - More than n vectors $\rightarrow$ linearly dependent.
 - n linearly independent vectors $\rightarrow$ a basis.

$$\mathbb{R}^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are basis of $\mathbb{R}^2$



Null space

The null space of A or the kernel of A denoted by $\mathcal{N}(A)$, is the set of real n -vectors x which satisfy

$$Ax = 0$$

that is,

$$\mathcal{N}(A) := \{x \in \mathbb{R}^n : Ax = 0\}$$

Let's compute a basis for the null space of the following matrix.

$$A = \begin{pmatrix} 0 & 2 & 2 & 4 & -2 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

reduced row echelon

$$A = \begin{pmatrix} 0 & 2 & 2 & 4 & -2 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

$r_1 \leftrightarrow r_2$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}$$

$r_3 - 2r_1$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & -4 & -4 & -8 & 4 \end{pmatrix}$$

$r_3 + 2r_2$
 $r_1 \rightarrow r_2$

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$r_2 / 2$

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$



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$$\hat{A} = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \hat{A}x = 0 \quad \text{nullity}(A) = 3$$

$$\begin{cases} x_1 + x_3 + x_5 = 0 \\ x_2 + x_3 + 2x_4 - x_5 = 0 \end{cases}$$

$$\rightarrow \begin{cases} x_1 = -x_3 - x_5 \\ x_2 = -x_3 - 2x_4 + x_5 \end{cases}$$

$$N(A) = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$N(A) = \text{span} \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$



Range

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \text{Range}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$\uparrow \quad \uparrow$
 $a_1 \quad a_2 = 2a_1$

$$\text{Rank}(A) = 1$$

The range of A , denoted by $\mathcal{R}(A)$, is the set of m -vectors of the form Ax where x is an n -vector, that is,

$$\mathcal{R}(A) := \{Ax : x \in \mathbb{R}^n\}$$

- The range of A is the set of vectors b for which the equation $Ax = b$ has a least one solution for x
- The rank of A is the *dimension of its range space*: $\text{rank } A := \dim \mathcal{R}(A)$

$$Ax = (a_1 \ a_2 \ \dots \ a_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$$

- The range of a matrix is the set of all linear combinations of its columns

Rank = Range 裡面有幾個 linear independent 的 vectors



Range

- Calculation of range basis and rank: use reduced row echelon
- $\text{rank } A = \text{maximum number of linearly independent columns of } A$
 $= \text{maximum number of linearly independent rows of } A$
- $\text{rank } A = \text{rank } A^T$

$$A = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 0 & 2 & 0 \end{pmatrix}$$

$$\text{rref}(A) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \leftarrow \text{rank}(A) = 2$$

$A \uparrow$

Coordinate transformation

- Suppose we have n scalar variables $x_1, x_2, \dots, x_n$ and we implicitly define new scalar variables $\xi_1, \xi_2, \dots, \xi_n$ by $x = \underline{U}\xi$, where U is an $n \times n$ invertible matrix
- x is the coordinate vector of itself with respect to the standard basis

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_x \leftarrow \underbrace{x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{e^1} + \underbrace{x_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{e^2} + \dots + \underbrace{x_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}}_{e^n} \rightarrow \text{basis}$$

- $x = U\xi$ can be written as $x = \xi_1 \underline{u_1} + \xi_2 \underline{u_2} + \dots + \xi_n \underline{u_n}$
- The vector ξ is the coordinate vector of the vector x wrt the new basis

Linear map

- Suppose we have a linear map from $\mathbb{R}^n$ into $\mathbb{R}^m$ defined by $y = Ax$
- The coordinate transformation $x = U\xi, y = V\eta$
- The map can also be described by $\eta = \Lambda\xi$ where $\Lambda = V^{-1}AU$

• **Useful result:** Suppose

$$Au^j = \alpha_{1j}v^1 + \alpha_{2j}v^2 + \dots + \alpha_{mj}v^m$$

Then the matrix $\Lambda = V^{-1}AU$ is uniquely given by

$$\Lambda_{ij} = \alpha_{ij}$$

that is Λ_{ij} is the i -th coordinate of the vector Au^j wrt the basis $v^1, v^2, \dots, v^m$, or, the j -th column of Λ is the coordinate vector of Au^j wrt the basis $v^1, v^2, \dots, v^m$.

$$\mathbb{R}^m \xleftarrow{m \times n} \mathbb{R}^n$$

$$V\eta = AU\xi$$

$$\eta = V^{-1}AU\xi = \Lambda\xi$$

$$\Rightarrow \overset{m \times n}{\Lambda} = \overset{1 \times m}{V^{-1}} \overset{m \times n}{A} \overset{n-1}{U}$$

$$V \Lambda = A U$$

$$V A U = A (u^1 \ u^2 \ \dots \ u^n) = (A u^1 \ A u^2 \ \dots \ A u^n)$$

$$V \Lambda = (v^1 \ v^2 \ \dots \ v^m) \begin{pmatrix} \Lambda_{11} & \dots & \Lambda_{1n} \\ \vdots & & \vdots \\ \Lambda_{m1} & \dots & \Lambda_{mn} \end{pmatrix}$$

$$V = \left(\Lambda_{11} v^1 + \dots + \Lambda_{m1} v^m \ \dots \ \underbrace{\Lambda_{j1} v^1 + \dots + \Lambda_{mj} v^m}_{j\text{th col.}} \right)$$



$\bar{j}$ -th column, $(1 \leq \bar{j} \leq n)$

$$\begin{aligned} \Lambda_{1\bar{j}} v^1 + \Lambda_{2\bar{j}} v^2 + \dots + \Lambda_{m\bar{j}} v^m &= A u^{\bar{j}} \\ &= \alpha_{1\bar{j}} v^1 + \dots + \alpha_{m\bar{j}} v^m \end{aligned}$$

$$\Rightarrow \underline{\Lambda_{i\bar{j}} = \alpha_{i\bar{j}}}$$



Example

Example 59 Suppose A is a 2×2 matrix and there are two linearly independent vectors u^1 and u^2 which satisfy:

$$Au^1 = 2u^1 \quad \text{and} \quad Au^2 = u^1 + 2u^2$$

$$\begin{aligned} \Lambda &= U^{-1} A U & \left. \begin{aligned} Au^1 &= \alpha_{11} u^1 + \alpha_{21} u^2 \\ Au^2 &= \alpha_{12} u^1 + \alpha_{22} u^2 \end{aligned} \right\} \\ &= \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} & \begin{aligned} \alpha_{11} &= 2 & \alpha_{21} &= 0 \\ \alpha_{12} &= 1 & \alpha_{22} &= 2 \end{aligned} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \end{aligned}$$



Behavior of LTI system

$$\dot{x} = A x \quad x: n\text{-vector}$$

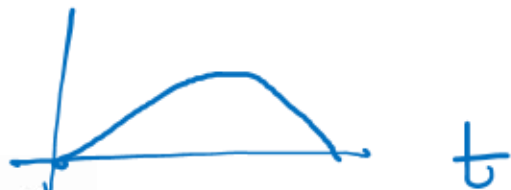
$$A: n \times n \text{ sys matrix}$$

Initial state x_0

$x(t) \leftarrow$ unique sol

* Time invariant = 時間不變, solution 不變

$x(t)$



$$t^* = t + 1$$

→



$x(t_0) = x_0$ 初始 state

↳ $\tilde{x}(0) = \hat{x}$

$$\dot{x} = Ax$$

$$x(t) = \tilde{x}(t - t_0) \quad \dot{\tilde{x}} = A \tilde{x}$$

$$x(t) = \Phi(t) x_0 \quad \Phi(t) : \text{state transition matrix}$$

$\Phi(t)$: 系統從初始狀態 x_0 演化到

時間 t 的狀態的轉移算子

(solution 對於 initial state 是 linear)



$$x[k+1] = Ax[k] \quad x[k] = n\text{-vectors}$$

$A = n \times n$ sys
matrix

$$x[k_0] = x_0 \quad k_0 = \text{initial time}$$

$x_0 = \text{initial state}$

$$x[k] = \tilde{x}[k - k_0] \quad (\text{Time Invariant})$$

$$k \geq k_0 \Rightarrow 0$$

$$\hookrightarrow x[k] = A^k x_0$$



$$k^* \geq 0$$

$$x(k^*) = A^{k^*} x_0$$

$$x[k] = A^{k-k_0} x[k_0]$$

$$k_0 \neq 0$$

$$x(k^*+1) = A \cdot x(k^*)$$

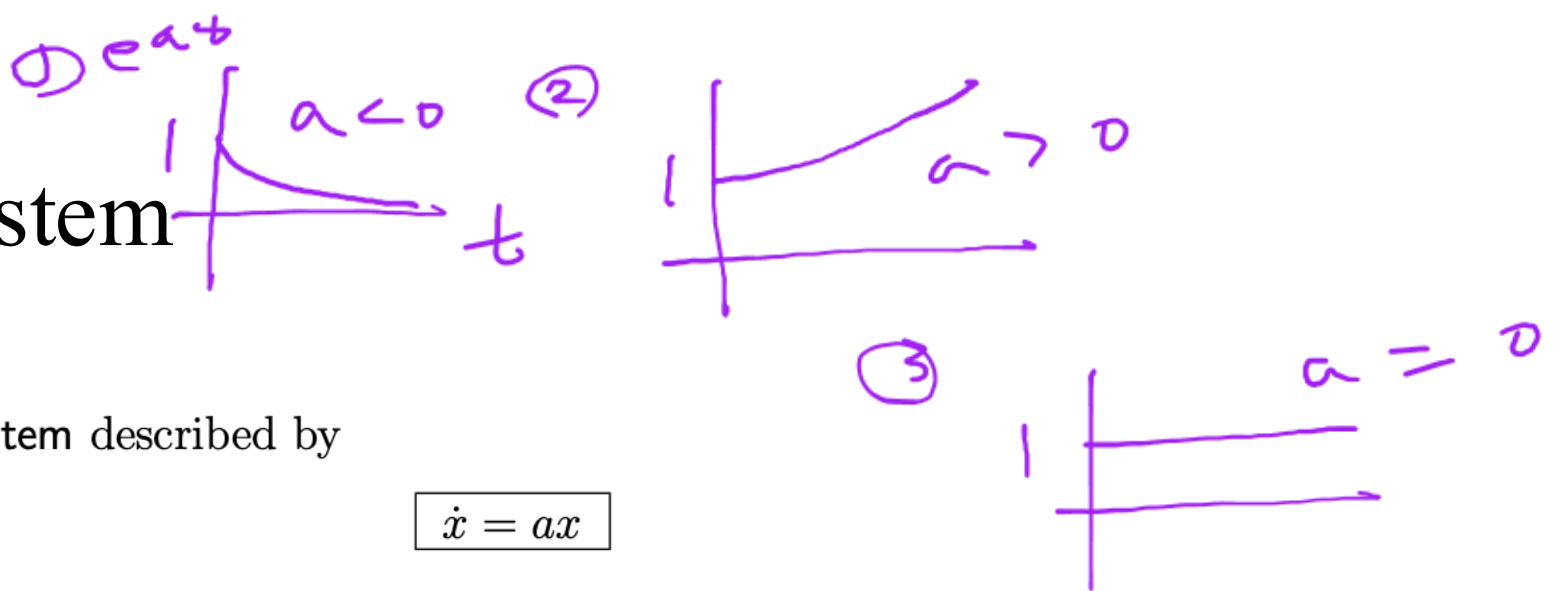
$$= A \cdot A^{k^*} x_0$$

$$= A^{(k^*+1)} x_0$$

$$x[k] = A^k x(0) \quad k_0 = 0$$



Behavior of LTI system



Consider a scalar system described by

$$\dot{x} = ax$$

where $x(t)$ and a are (real or complex) scalars. All solutions are given by:

$$x(t) = e^{at} x_0 ; x_0 = x(0)$$

↓

$$\dot{x} = a e^{at}$$

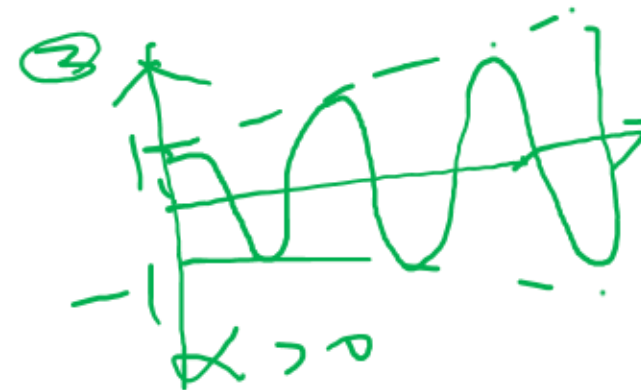
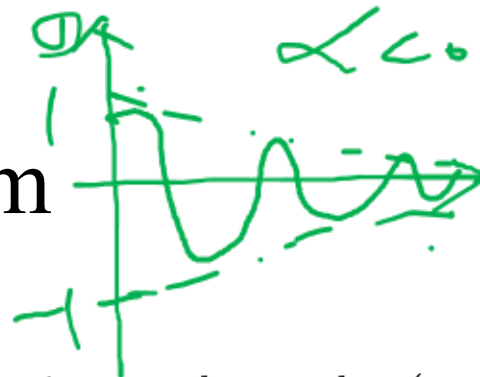
Real a

① $a < 0$, decay

② $a > 0$, grow

③ $a = 0$, stay the same

Behavior of LTI system



Complex a . Suppose a is a genuine complex number (unreal); that is,

$$a = \alpha + j\omega$$

where α and ω are real numbers. We call α the real part of a and denote this by $\alpha = \Re(a)$. Similarly, we call ω the imaginary part of a and denote this by $\omega = \Im(a)$.

$$e^{at} = e^{\alpha t + j\omega t} = \underbrace{e^{\alpha t}}_{\text{purple wavy line}} \underbrace{e^{j\omega t}}_{\text{blue wavy line}}$$

$$e^{j\omega t} = \cos \omega t + j \sin \omega t \rightarrow \text{purple wavy line}$$

$$|e^{j\omega t}| = 1 \quad \text{① } \Re(a) = \alpha < 0, \text{ decays}$$

$$\text{② } \alpha > 0, \text{ grows}$$

$$\text{③ } \alpha = 0, \text{ stay the same}$$



First order complex and second order real

Suppose a is a genuine complex number (unreal); that is,

$$a = \alpha + j\omega \quad \text{where} \quad \alpha = \Re(a), \quad \omega = \Im(a)$$

Since

$$x = \xi_1 + j\xi_2 \quad \text{where} \quad \xi_1 = \Re(x), \quad \xi_2 = \Im(x)$$

$$\begin{aligned} \bar{x} &= \overline{\alpha + j\omega} \overline{(\xi_1 + j\xi_2)} \\ &= \alpha \xi_1 - \omega \xi_2 + j(\omega \xi_1 + \alpha \xi_2) \end{aligned}$$
$$\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} \alpha & -\omega \\ \omega & \alpha \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}$$
$$\xi = \begin{bmatrix} \alpha & -\omega \\ \omega & \alpha \end{bmatrix} \xi$$

antisymmetric spacecraft

Discrete Time System

$$x(k+1) = ax(k)$$

where $x(k)$ and a are scalars. All solutions are given by:

$$x(k) = a^k x_0, \quad x_0 = x(0)$$

$$|x(k)| = |a^k x_0| = \underline{|a|}^k |x_0|$$

① $|a| < 1$, decay

② $|a| > 1$, grow

③ $|a| = 1$, magnitude remains the same

④ $a = 1$, stay the same, every state is equilibrium state



Eigenvalues and eigenvectors

DEFN. A complex number λ is an eigenvalue of A if there is a nonzero vector v such that

$$Av = \lambda v$$

The nonzero vector v is called an eigenvector of A corresponding to λ .

$$(\lambda I - A)v = 0$$

v is a non-zero vector $\Rightarrow \lambda I - A$ is singular

$$\Rightarrow \det(\lambda I - A) = 0$$



eigenvalues



$$* \det(\lambda I - A) = \alpha_n \lambda^n + \alpha_{n-1} \lambda^{n-1} + \dots + \alpha_1 \lambda + \alpha_0 \\ = \text{Charpoly}(A)$$

* A is $n \times n$, $\det(\lambda I - A)$ is a n -order polynomial
 $\Rightarrow$ 最多有 n 個 λ 的 eigenvalue

$$\det(sI - A) = \prod_{i=1}^n (s - \lambda_i)^{m_i}$$

* m_i is the algebraic multiplicity of λ_i

$$(s - \underline{2})^2 (s - 1) \quad \lambda_1 = 2 \rightarrow m_1 = 2 \\ \lambda_2 = 1 \rightarrow m_2 = 1$$



* eigenspace = $\{ \text{all } \vec{v} \mid A\vec{v} = \lambda\vec{v} \}$ + zero vector

* geometric multiplicity of λ

$\Rightarrow$ nullity of $\lambda I - A$

$\Rightarrow$ $\dim$ of linear independent $\vec{v}$ eigenvectors

$$1 \leq \text{GM}(\lambda) \leq \text{AM}(\lambda)$$

* A is invertible if and only if all eigenvalues are non zero



$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$$\det(sI - A) = \begin{vmatrix} s-3 & 1 \\ 1 & s-3 \end{vmatrix} = (s-3)^2 - 1$$

$$= s^2 - 6s + 8$$

$$= (s-2)(s-4)$$

$$\lambda_1 = 2, \lambda_2 = 4$$

$$(\lambda_i I - A) v = 0$$

$$\textcircled{1} \lambda_1 = 2$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0 \begin{cases} -v_1 + v_2 = 0 \\ v_1 - v_2 = 0 \end{cases}$$

$$v_1 = v_2 \Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$(2) \lambda = 4$$

$$(\lambda I - A)v = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} v = 0$$

$$v_1 \neq v_2 \Rightarrow \Rightarrow v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



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$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \quad \det(sI - A) = \det \begin{pmatrix} s-1 & -1 \\ 1 & s-1 \end{pmatrix}$$

$$= (s-1)^2 + 1$$

$$= s^2 - 2s + 2$$

$$\lambda_1 = 1 + j \quad \lambda_2 = 1 - j$$

$$\textcircled{D} (\lambda_1 I - A) v = \begin{pmatrix} j & -1 \\ 1 & j \end{pmatrix} v = 0$$

$$jv_1 - v_2 = 0 \quad \times -1$$

$$v_1 + jv_2 = 0 \quad \times -j$$

$$-jv_1 + v_2 = 0$$

$$v = \begin{pmatrix} 1 \\ j \end{pmatrix}$$



$$\textcircled{2} \lambda_2 = 1 - j$$

$$(\lambda_2 I - A) v = \begin{pmatrix} -j & -1 \\ 1 & -j \end{pmatrix} v = 0$$

$$\left. \begin{array}{l} -jv_1 - v_2 = 0 \\ v_1 - jv_2 = 0 \end{array} \right\} \begin{array}{l} jv_1 + v_2 = 0 \\ jv_1 + v_2 = 0 \end{array}$$

$$v = \begin{bmatrix} 1 \\ -j \end{bmatrix}$$



① eigenvalues are real numbers

$\Rightarrow$ eigenvectors are also real numbers

② complex eigenvalues $\Rightarrow$
complex eigenvectors

③ real & imaginary part of
a complex eigenvector are
linear independent

